

NPTEL Course on "Computer System Optimization for ML"

Lecture #49

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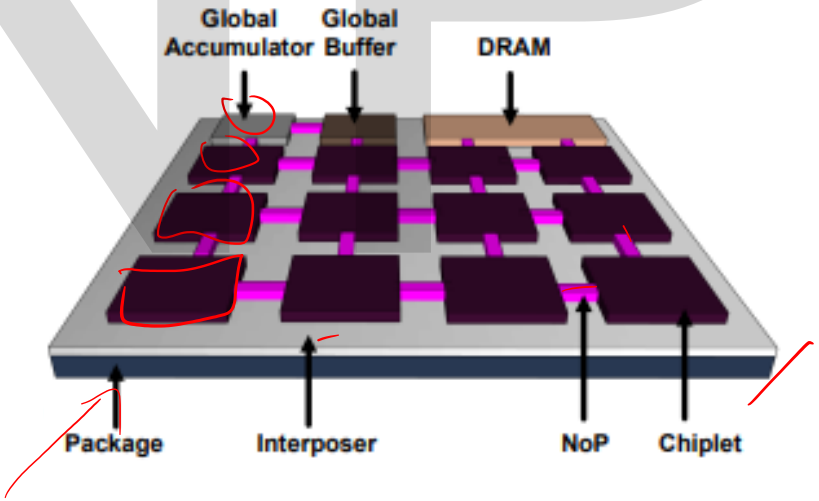
Agenda

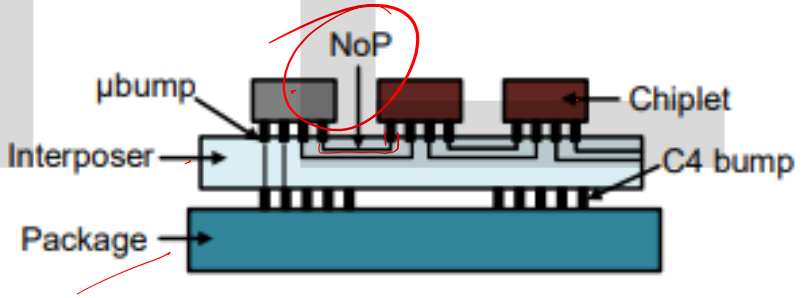
- **Chiplet-based Accelerator (contd.)**

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Background (2.5D Technology)

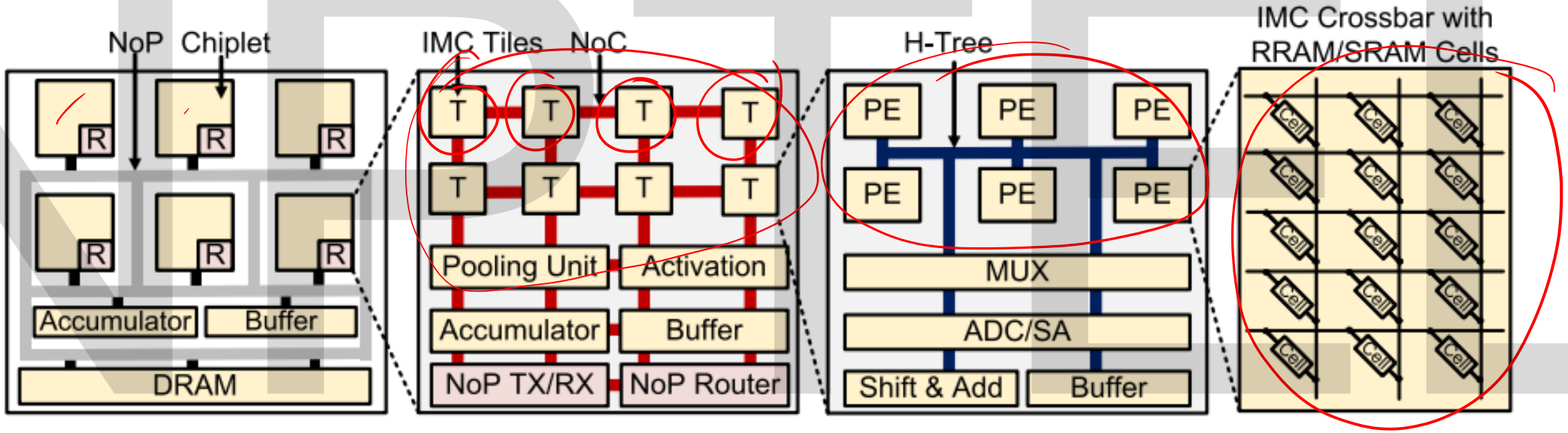
- Yield for monolithic IC is low
- 3D IC shows thermal hotspots due to increasing number of through-silicon vias (TSV)
- 2.5D technology has emerged recently which increases yield without much of





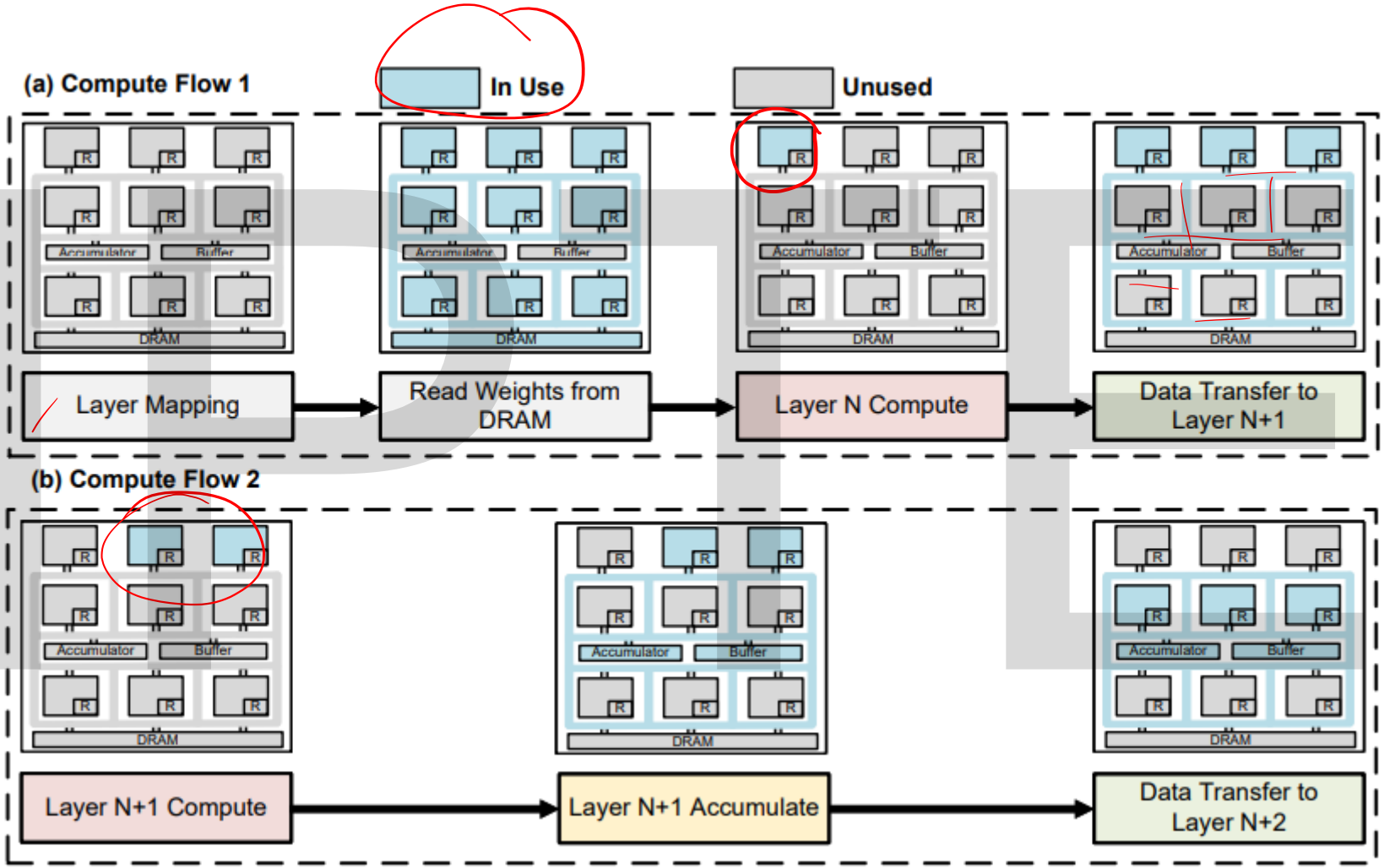
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Overview of the System



NDTL

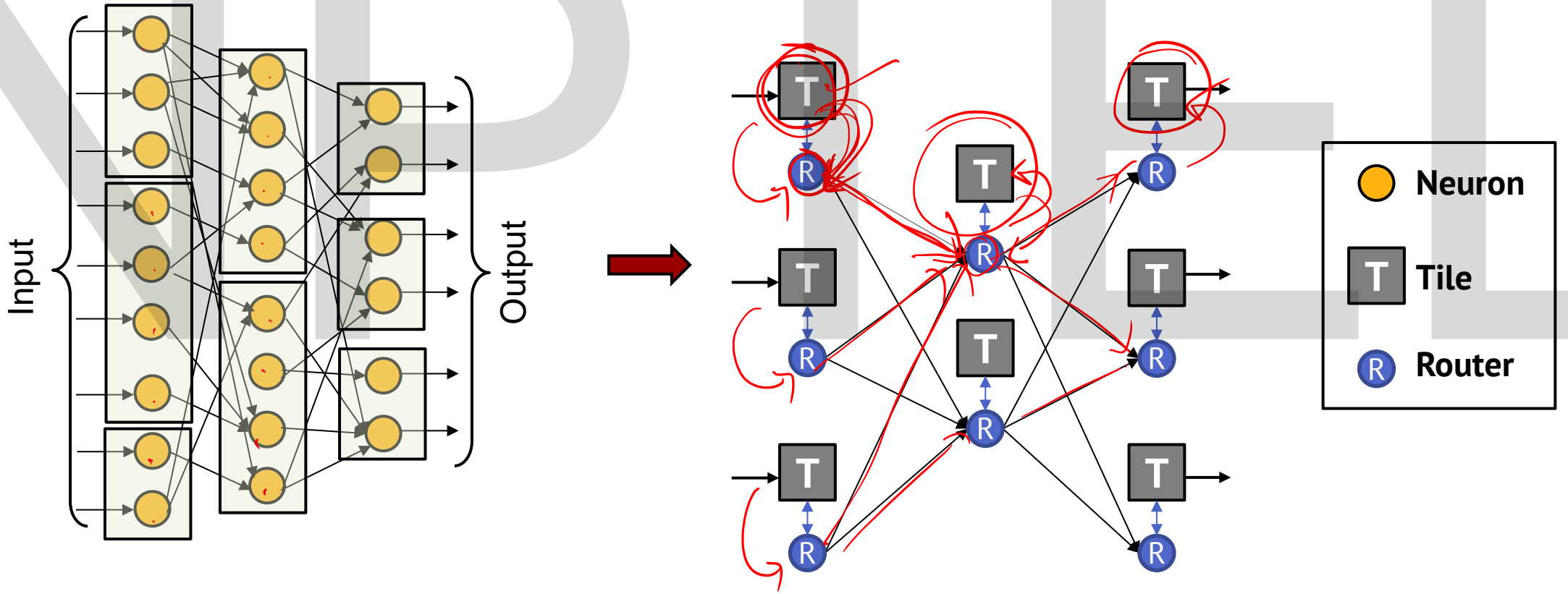
Dataflow



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Pictorial Representation

- A set of nodes are mapped to IMC tiles



Performance Estimation of Computation Elements

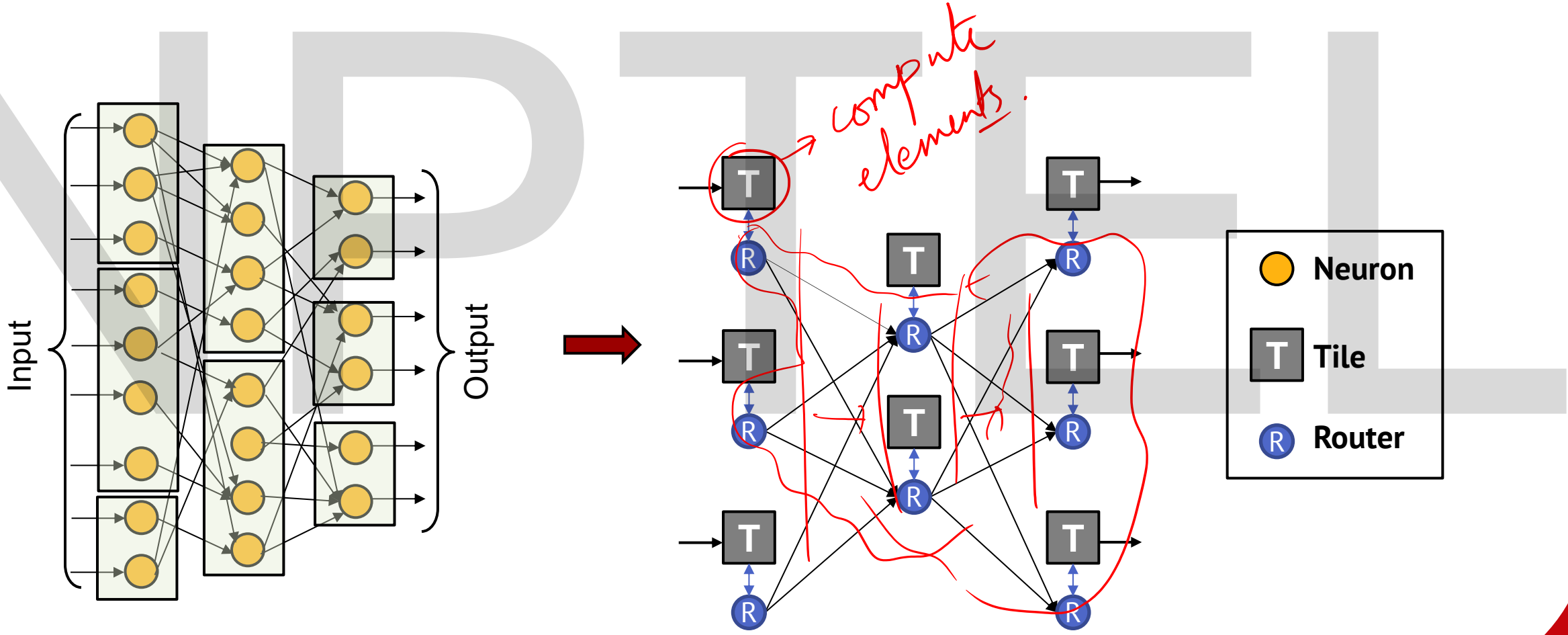
- **Modelling-based**
- **Analytical models for low-level circuit components via Predictive Technology Model (PTM) [3]**
  - Mux
  - Decoder
  - ADC/DAC
- **Required number of components are calculated**
  - Performance is estimated via scaling
- **Fast; since equations are executed again and again**

[3] Zhao, Wei, and Yu Cao. "Predictive technology model for nano-CMOS design exploration." ACM Journal on Emerging Technologies in Computing Systems (JETC) 3.1 (2007): 1-es.

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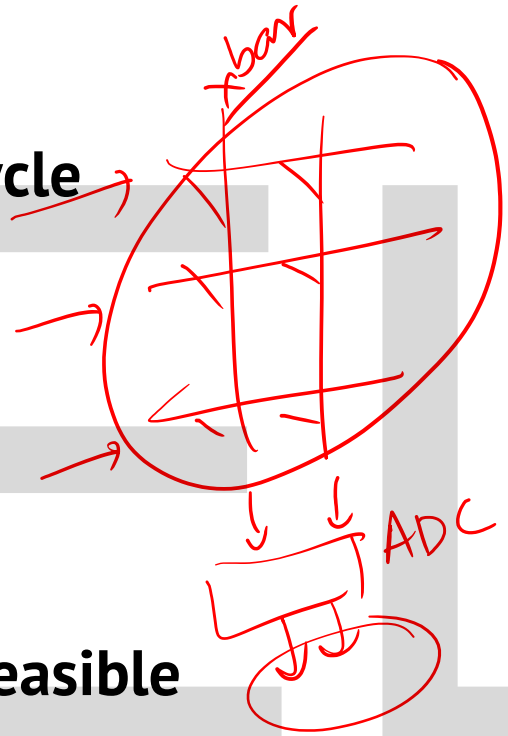
Pictorial Representation (Revisited)

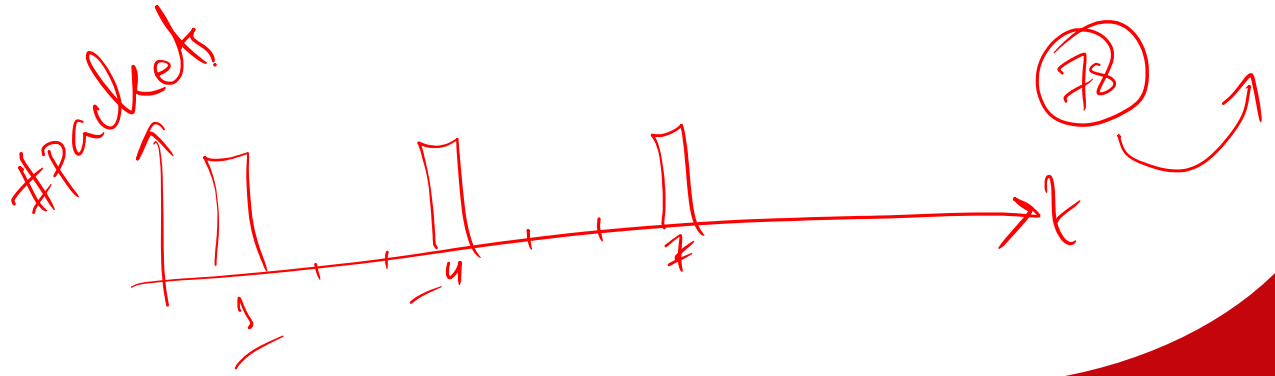
- A set of nodes are mapped to IMC tiles



Performance Estimation of Communication

- **Typically, a block of computation is performed at once**
- **Many packets are ready to be communicated at a single cycle**
  - No particular distribution in arrival rate
- **Need to know the total latency**
  - Time taken by the communication to be finished
  - Average latency is not important
- **Deploying queuing theory-based analytical model is not feasible**
- **Cycle accurate simulation is performed to estimate communication performance**





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Step 1: Trace File Generation

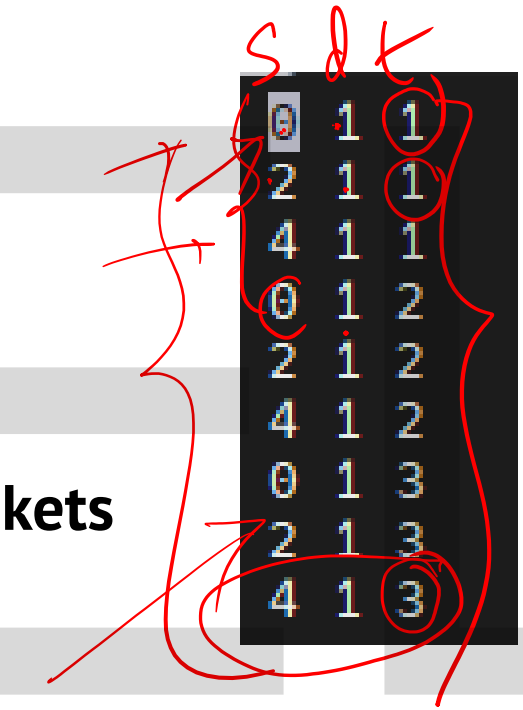
- **Trace file contains below information**
  - **Source**
  - **Destination**
  - **Time stamp**

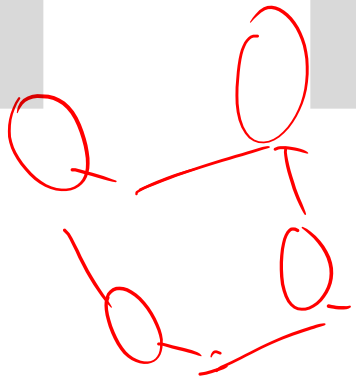
Algorithm 2: NoC (or NoP) Trace Generation

Input: Number of tiles for each layer (for each chiplet in case of NoP) ($|\mathcal{T}|$), Number of input activations for each layer(A), Number of chiplets (C), Layer to chiplet mapping ($\mathcal{L} \rightarrow C$), Quantization bit ($Q$), Bus width ($W$)
Output: Trace file for each chiplet (tr^c)
for $c = 1 : |C|$ **do**
  Find index of the first layer (L_c^S) and the last layer (L_c^E) in the chiplet from $\mathcal{L} \rightarrow C$
  for $l = L_c^S : L_c^E$ **do**
    $k = 0$ /* Initialize timestamp */
    Find index of first source tile (T_l^S) and last source tile (T_l^E)
    Find index of first destination tile (T_{l+}^S) and last destination tile (T_{l+}^E)
    $N_p = \lceil \frac{A(l)Q}{W} \rceil$ /* Number of packets */
    for $n = 1 : N_p$ **do**
      for $s = T_l^S : T_l^E$ **do**
        for $d = T_{l+}^S : T_{l+}^E$ **do**
          $tr^c \leftarrow [tr^c; [s, d, k]]$
          $k \leftarrow k + 1$ /* Increment timestamp */
        end
        $k \leftarrow k + 1$ /* Increment timestamp */

Example of a Trace File

- Three columns
  - Source router
  - Destination router
  - Time stamp
- Routers cannot send more than one packet at once cycle
- Number of lines in the trace file represents number of packets





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Simulating the Trace File

- Cycle accurate simulation
  - BookSim
- Mesh interconnect
- The timestamp when all the packets are received is collected
- BookSim also provides area and energy estimation

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THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)